

Question 1 [13 points]

EcoClean is a company specializing in the production of eco-friendly cleaning agents. Their products are produced by blending various organic raw materials, each sourced from different suppliers, to create a series of blends that can be sold to retail and industrial clients alike. Due to environmental concerns and regulations, each cleaning agent (referred to as a “blend”) must meet specific quality standards while each supplier has limited raw material availability. EcoClean is committed to minimizing production costs while ensuring that all customer demands for their cleaning agents are exactly met and that the amount of each raw material used across all blends does not exceed their availability. The company produces 50 unique blends each made from a combination of 100 organic raw materials. Each raw material has its own availability, cost, and max proportion parameter, $p_j^{\max}$ (see *blending_materials.csv* for more details) for $j = 1, \dots, 100$. To maintain product quality and ensure customer satisfaction, each blend must meet certain quality constraints, and thus, the proportions of specific raw materials in each blend are strictly regulated. To facilitate this, customer demand for each blend (tons), the min/max quality level (i.e., $Q_i^{\min}$ and $Q_i^{\max}$) of each cleaning agent, and the per ton quality contribution of raw material $j = 1, \dots, 100$ to blend $i = 1, \dots, 50$ are provided in *blending_blends.csv*. Furthermore, two operational restrictions must be adhered to:

1. The average quality of each blend must remain within a specified range $[Q_i^{\min}, Q_i^{\max}]$ for all i .
2. The amount of raw material j that can be used in any blend i cannot exceed a certain proportion $p_j^{\max}$ of all the raw materials that are used to create that blend.

Formulate and solve a linear program providing a minimum-cost production plan that (i) meets all demand; (ii) satisfies the operational requirements; (iii) and does not exceed raw material availability.

- (a) What do the decision variables represent and how many of them are there?
- (b) What is the minimum production cost associated with this linear program?
- (c) If the supplier of raw material $j = 5$ reduced its price by \$1.00 per ton, would this material now be included in any blend formulations? Please explain your logic and provide evidence.
- (d) How sensitive is the solution to changes in the minimum average quality constraint? That is, if $Q_i^{\min}$ were increased by one unit for each blend $i = 1, \dots, 50$, would the optimal solution change? Please provide quantitative justification for your response.
- (e) Assume EcoGreen has the opportunity to purchase 20 additional tons of raw material $j = 73$. What is the maximum amount these additional units could be priced at for EcoGreen to still achieve a net reduction in costs? Please justify your response.
- (f) Suppose that the operational constraint #2 is altered such that it now reads as follows: *The amount of raw material j that can be used in any blend i cannot exceed a certain proportion $p_j^{\max}$ of the sum of the squared values of each raw material used for that blend.* Given this updated model, what is the minimum production cost for the nonlinear program?
- (g) *Instead* of modifying operational constraint #2, assume that the constraint set is linear but the objective is now to minimize the sum of squared production costs, i.e., the amounts used in each of the blends are squared. What is the optimal value for the objective function in this case?
- (h) How many non-zero decision variables are there in parts (f) and (g)? Why does this occur?
- (i) Ridge regression encourages sparse models by shrinking many coefficients towards zero. However, the opposite appears to be happening in part (g). Explain what is causing this difference.

Question 2 [13 points]

Atlas Construction, a large-scale infrastructure company, is preparing to hire a group of welders for a high-priority construction job. The project requires a team of 16 welders with various skill sets and experience. Each welder has been evaluated based on their proficiency in *four* key welding techniques: Shielded Metal Arc Welding (SMAW), Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), and Gas Tungsten Arc Welding (GTAW). In addition, welders have been assessed for their safety record, speed, and experience. For their safety record and speed, each welder has received a rating from 1 to 4, with 4 being excellent, 3 being above average, 2 being adequate, and 1 being below average. For experience, all that has been noted is whether each welder has been certified for more or less than 10 years. Atlas Construction has received 144 applicants and wishes to ensure that the team they hire has a balanced set of skills, while also adhering to several project constraints.

- At least 50% of the hired welders must be proficient in both SMAW and GMAW.
- For each of the welding techniques, there should be least two welders that are proficient.
- At least 30% of the hired welders must have more than 10 years of experience.
- The average rating of all hired welders must be no less than 3.9 for safety and 3.1 for speed.
- There should be at least three welders with a speed rating of 1.0 *and* a safety rating of 3 or 4.
- The number of welders hired from the range 70 to 100 must be at least one more than twice the number of welders that are hired from the range 101 to 130.

Using the information contained in the file *welders_data.csv*, formulate and solve a binary optimization problem to determine which of the 144 welders should be hired so as to maximize the *average speed of the team* while ensuring that all of the above constraints are satisfied.

- (a) What do the decision variables represent and how many of them are there?
- (b) What is the optimal objective function value of the binary program and its linear relaxation where all binary decision variables can take on continuous values between 0 and 1?
- (c) As observed from part (b), the optimal objective function values of the binary program and its relaxation are not equal. What does this imply? If they were equal, what would it indicate about the computational approach to solving this optimization model?
- (d) If the constraint requiring *at least three welders with a speed rating of 1.0 and a safety rating of 3 or 4* is removed, what is the new optimal objective function value, and why does it change?
- (e) Are there any redundant constraints? If so, identify the constraint(s) and explain why they are redundant. If not, explain why there are no redundant constraints.
- (f) How many solutions have the same optimal objective value as the binary program in part (b)?
- (g) From a managerial standpoint, what is the biggest drawback of making hiring decisions based solely on an individual's own qualities when assembling a team for a project like this?
- (h) Suppose we use a specialized AI tool to estimate the speed of welder i (i.e., v_{ii}) and the additional speed gained (lost) when working alongside welder j (i.e., v_{ij}), value which are provided in the file *welders_speed_data.csv*. What is the new objective function (use mathematical notation) and what is its value after optimally solving the corresponding nonlinear optimization problem?

- (i) Can this nonlinear binary program be reformulated as a linear program? Please provide a mathematical description of the transformation but do not implement it in your code.
- (j) By considering qualities associated hiring two team members together, this approach is potentially more realistic than making hiring decisions based solely on an individual's qualities. However, some issues remain given that we want to hire 16 welders. Briefly explain.
- (k) Suppose that, to ameliorate the drawbacks in part (g) and part (j), we create a binary program where the decision variable is 1 if a particular group of 16 welders is selected and 0 otherwise. Modeling the problem in this way would capture all possible group interactions. How many decision variables would this model require? Do you think it can be optimally solved? What is a potential drawback of using this solution approach when making hiring decisions?